

Assignment 6

Question 1 :

A concentrated load of 2500 kN is applied at the ground surface. Determine the vertical stress at a point up to 6 m depth and at a horizontal distance of 5 m from the axis of the load. Plot the graph by using Excel. Use Boussinesq's Theory.

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# Calculating the stress by Boussinesq's Theory
Q = int(input("Enter the value of given load: "))
z = int(input("Enter the distance of vertical stress: "))
r = int(input("Enter the distance of horizontal stress: "))
stress = (3 * Q * (1 / (1 + (r/z)**2))**2.5) / (2 * 3.14 * (z**2))
print("The value of stress is:", stress)
```

Output:

Enter the value of given load :2500

Enter the distance of vertical stress :6

Enter the distance of horizontal stress :5

The value of stress is: 8.876275703713446

r = Radial Distance	Stress
0.25	33.030549
0.5	32.605085
0.75	31.912855
1	30.977834
1.25	29.831091
1.5	28.5086579
1.75	27.0492535
2	25.492087
2.25	23.8749188
2.5	22.2325147
2.75	20.595549
3	18.9899616
3.25	17.4367141

3.5	15.9518874
3.75	14.54702
4	13.2296091
4.25	12.003698
4.5	10.8704883
4.75	9.82393245
5	8.8762757
5.25	8.00853255
5.5	7.22088892
5.75	6.50803129
6	5.86440735
6.25	5.28442586
6.5	4.76260436
6.75	4.29367381
7	3.87264829
7.25	3.49486739
7.5	3.15601773
7.75	2.85213881
8	2.57961783

